

```
from google.colab import files
uploaded = files.upload()
```

[Choose Files](#) Screenshot... 214117.png

Screenshot 2026-07-23 214117.png(image/png) - 288815 bytes, last modified: 7/23/2026 - 100% done
Saving Screenshot 2026-07-23 214117.png to Screenshot 2026-07-23 214117.png

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load a grayscale image (Replace with your medical image)
image = cv2.imread("Screenshot 2026-07-23 214117.png", cv2.IMREAD_GRAYSCALE)

if image is None:
    print("Error: Image not found!")
    # It's better to raise an error or return than to call exit() in a Colab cell
    raise FileNotFoundError("The specified image file was not found. Please ensure")

# -----
# Pixel Resolution Demonstration
# -----
small = cv2.resize(image, (64, 64), interpolation=cv2.INTER_LINEAR)
pixel_low = cv2.resize(small, image.shape[:,-1], interpolation=cv2.INTER_NEAREST)

# -----
# Intensity Resolution Demonstration
# -----
def reduce_gray_levels(img, levels):
    factor = 256 // levels
    reduced = (img // factor) * factor
    return reduced

intensity_16 = reduce_gray_levels(image, 16) # 4-bit
```

```
intensity_4 = reduce_gray_levels(image, 4)      # 2-bit

# -----
# Display Results
# -----
plt.figure(figsize=(12,8))

plt.subplot(2,2,1)
plt.imshow(image, cmap='gray')
plt.title("Original Image")
plt.axis("off")

plt.subplot(2,2,2)
plt.imshow(pixel_low, cmap='gray')
plt.title("Low Pixel Resolution")
plt.axis("off")

plt.subplot(2,2,3)
plt.imshow(intensity_16, cmap='gray')
plt.title("16 Gray Levels (4-bit)")
plt.axis("off")

plt.subplot(2,2,4)
plt.imshow(intensity_4, cmap='gray')
plt.title("4 Gray Levels (2-bit)")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Original Image



Low Pixel Resolution



16 Gray Levels (4-bit)



4 Gray Levels (2-bit)



